# CareLens AI — Product Documentation

CareLens AI is a local web-based symptom-monitoring and preliminary health-guidance assistant. It combines hybrid retrieval (FAISS semantic search plus BM25 lexical search), cross-encoder reranking, and grounded local text generation with a software safety layer. All data stays on the user's machine in SQLite.

# 1. Product goals

Provide understandable preliminary health information, show supporting evidence and match relevance, escalate emergencies clearly, and keep health data private and local.

# 2. Key features

Secure accounts; conversational symptom input; top matching conditions; contributing-symptom evidence; health guidance; match-relevance indicator; emergency safety notices; consultation history with search, detail view and deletion.

# 3. User workflow

Register, sign in, describe symptoms, review the grounded response with evidence cards and safety advice, revisit or delete past consultations, sign out.

# 4. System architecture

USER SYMPTOMS -> preprocessing -> HyDE-style local expansion -> embeddings -> hybrid retrieval (FAISS + BM25) -> score fusion (0.65 semantic / 0.35 lexical, configurable) -> cross-encoder rerank (top-5) -> grounded generation -> safety filter -> UI + SQLite storage.

# 5. Technology stack

Python, FastAPI, Uvicorn, SQLite, SQLAlchemy, Pydantic, Jinja2, Sentence Transformers (all-MiniLM-L6-v2), FAISS CPU, rank-bm25, cross-encoder ms-marco-MiniLM-L-6-v2, Flan-T5 small with evidence-based fallback, scikit-learn, pandas, numpy.

# 6. Dataset and preprocessing

Normalized knowledge-base records with record_id, disease, symptoms, description, guidance, severity, red_flags, source. Text normalized, duplicates removed, symptom separators standardized, canonical retrieval text built as 'Disease: ... Symptoms: ... Description: ... Guidance: ...', validated and saved to data/processed/.

# 7. Hybrid retrieval

Independent FAISS semantic path and BM25 lexical path, min-max normalized and fused with configurable weights. Candidate pool of ~15, reranked to top-5.

# 8. Query expansion

Rule-based local HyDE-style step: normalizes the query and constructs a concise hypothetical medical-information representation ('Medical symptom presentation involving ...') used alongside the original query. No external API, no separate trained model.

# 9. FAISS / BM25 / reranking

FAISS IndexFlatIP over L2-normalized embeddings (cosine-equivalent). BM25Okapi over tokenized retrieval texts. Cross-encoder ms-marco-MiniLM-L-6-v2 reranks candidates; falls back to hybrid order offline. Raw scores are translated to a simple match-relevance display.

# 10. Response generation & safety

Flan-T5 answers only from retrieved context with instructions to avoid definitive diagnosis and prescriptions, note uncertainty, and recommend professional care. A deterministic evidence-based fallback assembles the same content if generation is unavailable. A rule layer flags emergency patterns (chest pain, breathing difficulty, bleeding, unconsciousness, stroke-like signs, anaphylaxis, self-harm, seizures) and always shows a non-diagnosis disclaimer.

# 11. Authentication & database

SQLite via SQLAlchemy; bcrypt password hashing; signed token sessions in HttpOnly cookies; per-user ownership enforced on all history endpoints; input validation throughout.

# 12. How to run

Double-click run.bat (recommended), or use Docker (run-docker.bat), or follow HOW_TO_RUN.md Method C for manual setup. The app binds to 127.0.0.1 and opens http://localhost:8000.

# 13. Limitations & responsible use

Preliminary information only, not a diagnosis; knowledge base is limited; similarity is not medical probability; never prescribe medication. In emergencies seek immediate professional care.

# 14. Evaluation methodology and measured results

Evaluation runs on a held-out record per condition (one random record per disease), querying with its symptom list and mapping the top-ranked retrieved disease to the expected label. Metrics below are the actual outputs of scripts/evaluate.py.

n_test: 30

top1_accuracy: 1.0

top3_accuracy: 1.0

recall_at_3: 1.0

macro_precision: 1.0

macro_recall: 1.0

macro_f1: 1.0

mrr_at_3: 1.0

mean_latency_ms: 245.18

embedding_mode: sentence-transformer

reranker_available: True

generator_available: True